Laboratory

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Example 1. *Let $q = 3$ and $r = 5$; the characteristic of $\mathbb{F}_3$ does not divide r. Over $\mathbb{F}_3$, let $P(0) = \mathrm{diag}\{1, 2, 2, 1, 2\}$. Let*

$$H_1 = \begin{bmatrix} P(0) & P(2) & P(4) & P(1) & P(3) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(1) & P(2) & P(3) & P(4) \end{bmatrix}.$$

Then $\mathrm{rank}_q(H_1) = 5$, $\mathrm{rank}_q(H_2) = 5$ and $\mathrm{rank}_q(H_1 H_2^T) = 1$. There exists a non-binary EA-QC-QLDPC code with parameters $[[25, 16; 1]]_3$, whose minimum distance is 2.

Example 2. *Let $q = 3$ and $r = 5$; the characteristic of $\mathbb{F}_3$ does not divide r. Over $\mathbb{F}_3$, let $P(0) = \mathrm{diag}\{2, 1, 1, 2, 1\}$. Let*

$$H_1 = \begin{bmatrix} P(0) & P(0) & P(0) & P(0) & P(0) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(3) & P(4) & P(2) & P(1) \\ P(0) & P(4) & P(2) & P(1) & P(3) \end{bmatrix}.$$

Then $\mathrm{rank}_q(H_1) = 5$, $\mathrm{rank}_q(H_2) = 9$ and $\mathrm{rank}_q(H_1 H_2^T) = 1$. There exists a non-binary EA-QC-QLDPC code with parameters $[[25, 12; 1]]_3$, whose minimum distance is 2.

Example 3. *Let $q = 3$ and $r = 5$; the characteristic of $\mathbb{F}_3$ does not divide r. Over $\mathbb{F}_3$, let $P(0) = \mathrm{diag}\{1, 1, 2, 2, 1\}$. Let*

$$H_1 = \begin{bmatrix} P(0) & P(0) & P(0) & P(0) & P(0) \\ P(0) & P(3) & P(1) & P(4) & P(2) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(1) & P(2) & P(3) & P(4) \\ P(0) & P(4) & P(3) & P(2) & P(1) \end{bmatrix}.$$

Then $\mathrm{rank}_q(H_1) = 9$, $\mathrm{rank}_q(H_2) = 9$ and $\mathrm{rank}_q(H_1 H_2^T) = 1$. There exists a non-binary EA-QC-QLDPC code with parameters $[[25, 8; 1]]_3$, whose minimum distance is 4.

Example 4. *Let $q = 3$ and $r = 5$; the characteristic of $\mathbb{F}_3$ does not divide r. Over $\mathbb{F}_3$, let $P(0) = \mathrm{diag}\{2, 2, 1, 1, 2\}$. Let*

$$H_1 = \begin{bmatrix} P(0) & P(1) & P(2) & P(3) & P(4) \\ P(0) & P(4) & P(3) & P(2) & P(1) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(0) & P(0) & P(0) & P(0) \\ P(0) & P(2) & P(4) & P(1) & P(3) \\ P(0) & P(3) & P(1) & P(4) & P(2) \end{bmatrix}.$$

Then $\mathrm{rank}_q(H_1) = 9$, $\mathrm{rank}_q(H_2) = 13$ and $\mathrm{rank}_q(H_1 H_2^T) = 1$. There exists a non-binary EA-QC-QLDPC code with parameters $[[25, 4; 1]]_3$, whose minimum distance is 4.

Example 5. *Let $q = 3$ and $r = 7$; the characteristic of $\mathbb{F}_3$ does not divide r. Over $\mathbb{F}_3$, let $P(0) = \mathrm{diag}\{1, 2, 1, 1, 2, 2, 1\}$. Let*

$$H_1 = \begin{bmatrix} P(0) & P(6) & P(5) & P(4) & P(3) & P(2) & P(1) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(1) & P(2) & P(3) & P(4) & P(5) & P(6) \end{bmatrix}.$$

Then $\mathrm{rank}_q(H_1) = 7$, $\mathrm{rank}_q(H_2) = 7$ and $\mathrm{rank}_q(H_1 H_2^T) = 1$. There exists a non-binary EA-QC-QLDPC code with parameters $[[49, 36; 1]]_3$, whose minimum distance is 2.

Example 6. *Let $q = 3$ and $r = 7$; the characteristic of $\mathbb{F}_3$ does not divide r. Over $\mathbb{F}_3$, let $P(0) = \mathrm{diag}\{2, 1, 2, 1, 1, 2, 2\}$. Let*

$$H_1 = \begin{bmatrix} P(0) & P(1) & P(2) & P(3) & P(4) & P(5) & P(6) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(5) & P(3) & P(1) & P(6) & P(4) & P(2) \\ P(0) & P(6) & P(5) & P(4) & P(3) & P(2) & P(1) \end{bmatrix}.$$

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Then $\mathrm{rank}_q(H_1) = 7$, $\mathrm{rank}_q(H_2) = 13$ *and* $\mathrm{rank}_q(H_1 H_2^T) = 1$. *There exists a non-binary EA-QC-QLDPC code with parameters* $[[49, 30; 1]]_3$, *whose minimum distance is* 2.

Example 7. *Let* $q = 3$ *and* $r = 7$; *the characteristic of* $\mathbb{F}_3$ *does not divide* r. *Over* $\mathbb{F}_3$, *let* $P(0) = \mathrm{diag}\{1, 2, 2, 1, 2, 1, 1\}$. *Let*

$$H_1 = \begin{bmatrix} P(0) & P(0) & P(0) & P(0) & P(0) & P(0) & P(0) \\ P(0) & P(1) & P(2) & P(3) & P(4) & P(5) & P(6) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(2) & P(4) & P(6) & P(1) & P(3) & P(5) \\ P(0) & P(6) & P(5) & P(4) & P(3) & P(2) & P(1) \end{bmatrix}.$$

Then $\mathrm{rank}_q(H_1) = 13$, $\mathrm{rank}_q(H_2) = 13$ *and* $\mathrm{rank}_q(H_1 H_2^T) = 1$. *There exists a non-binary EA-QC-QLDPC code with parameters* $[[49, 24; 1]]_3$, *whose minimum distance is* 4.

Example 8. *Let* $q = 3$ *and* $r = 7$; *the characteristic of* $\mathbb{F}_3$ *does not divide* r. *Over* $\mathbb{F}_3$, *let* $P(0) = \mathrm{diag}\{1, 2, 1, 2, 2, 1, 2\}$. *Let*

$$H_1 = \begin{bmatrix} P(0) & P(2) & P(4) & P(6) & P(1) & P(3) & P(5) \\ P(0) & P(6) & P(5) & P(4) & P(3) & P(2) & P(1) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(0) & P(0) & P(0) & P(0) & P(0) & P(0) \\ P(0) & P(3) & P(6) & P(2) & P(5) & P(1) & P(4) \\ P(0) & P(5) & P(3) & P(1) & P(6) & P(4) & P(2) \end{bmatrix}.$$

Then $\mathrm{rank}_q(H_1) = 13$, $\mathrm{rank}_q(H_2) = 19$ *and* $\mathrm{rank}_q(H_1 H_2^T) = 1$. *There exists a non-binary EA-QC-QLDPC code with parameters* $[[49, 18; 1]]_3$, *whose minimum distance is* 4.

Example 9. *Let* $q = 3$ *and* $r = 7$; *the characteristic of* $\mathbb{F}_3$ *does not divide* r. *Over* $\mathbb{F}_3$, *let* $P(0) = \mathrm{diag}\{2, 1, 1, 2, 1, 2, 1\}$. *Let*

$$H_1 = \begin{bmatrix} P(0) & P(0) & P(0) & P(0) & P(0) & P(0) & P(0) \\ P(0) & P(6) & P(2) & P(3) & P(5) & P(4) & P(1) \\ P(0) & P(5) & P(4) & P(6) & P(3) & P(1) & P(2) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(3) & P(1) & P(5) & P(6) & P(2) & P(4) \\ P(0) & P(2) & P(3) & P(1) & P(4) & P(6) & P(5) \\ P(0) & P(4) & P(6) & P(2) & P(1) & P(5) & P(3) \end{bmatrix}.$$

Then $\mathrm{rank}_q(H_1) = 19$, $\mathrm{rank}_q(H_2) = 19$ *and* $\mathrm{rank}_q(H_1 H_2^T) = 1$. *There exists a non-binary EA-QC-QLDPC code with parameters* $[[49, 12; 1]]_3$, *whose minimum distance is* 6.

Example 10. *Let* $q = 5$ *and* $r = 3$; *the characteristic of* $\mathbb{F}_5$ *does not divide* r. *Over* $\mathbb{F}_5$, *let* $P(0) = \mathrm{diag}\{1, 2, 4\}$. *Let*

$$H_1 = \begin{bmatrix} P(0) & P(2) & P(1) \end{bmatrix},$$

$$H_2 = \begin{bmatrix} P(0) & P(1) & P(2) \end{bmatrix}.$$

Then $\mathrm{rank}_q(H_1) = 3$, $\mathrm{rank}_q(H_2) = 3$ *and* $\mathrm{rank}_q(H_1 H_2^T) = 1$. *There exists a non-binary EA-QC-QLDPC code with parameters* $[[9, 4; 1]]_5$, *whose minimum distance is* 2.